Exploratory Data Analysis (EDA) Summary Report

1. Introduction

This report provides a clear overview of data quality, uncovers key risk factors, and informs strategies for enhancing delinquency predictions and business decisions.

2. Dataset Overview

This section summarizes the dataset, including the number of records, key variables, and data types. It also highlights any anomalies, duplicates, or inconsistencies observed during the initial review.

Key dataset attributes:

- Number of records: 500 Customers

- Key variables:

S.No.	Numeric Fields	Categorical Fields
1.	*Age* (years)	*Customer_ID* (unique ID)
2.	*Income* (annual dollars)	*Employment_Status* (e.g. “Employed”, “Self-employed”, “Unemployed”, “Retired”, with inconsistent labeling like “EMP” vs “Employed”)
3.	*Credit_Score* (300–850)	*Credit_Card_Type* (e.g. Student, Standard, Platinum, Gold, Business)
4.	*Credit_Utilization* (fraction of credit used)	*Location* (city)
5.	*Missed_Payments* (count of missed payments)	payment status for each of six months (*Month_1 - Month_6*, values “On-time”, “Late”, “Missed”)
6.	*Loan_Balance* (dollar balance)	
7.	*Debt_to_Income_Ratio* (debt/income fraction)	
8.	*Account_Tenure* (years).	

- Data types:
  Mostly numerical (Age, Income, Credit_Score, Credit_Utilization, Missed_Payments, Delinquent_Account (0/1 flag), Loan_Balance, Debt_to_Income_Ratio, Account_Tenure) and categorical (Employment_Status, Credit_Card_Type, Location, Month_1–Month_6). The monthly status fields are ordinal categorical.

3. Missing Data Analysis

Identifying and addressing missing data is critical to ensuring model accuracy. This section outlines missing values in the dataset, the approach taken to handle them, and justifications for the chosen method.

Key missing data findings:

- Variables with missing values:
  1. *Income* – several records missing
  2. *Loan_Balance* – some records missing
  3. Credit_Score – a few records missing
  4. Month_1 to Month_6 – occasional missing payment status

Column	*Missing Data Handling*	*Justification*
Income	Median imputation	Median is robust to outliers and maintains typical income distribution.
Loan_Balance	Median imputation	Preserves central tendency without biasing models with extreme values.
Credit_Score	Median imputation	Ensures numeric continuity for modeling delinquency risk
Month_1 to Month_6	"Missing/Unknown" category	Retains information about payment behavior while handling gaps in records

4. Key Findings and Risk Indicators

This section identifies trends and patterns that may indicate risk factors for delinquency. Feature relationships and statistical correlations are explored to uncover insights relevant to predictive modeling.

Key findings:

- Correlations observed between key variables:
  <u>1. Credit Score & Delinquency:</u> Strong negative correlation; lower credit scores are associated with higher delinquency risk.
  <u>2. Debt-to-Income Ratio & Missed Payments:</u> Positive correlation; higher ratios indicate higher likelihood of missed payments.
  <u>3. Credit Utilization & Recent Delinquency:</u> Moderate positive correlation; customers using a large portion of available credit tend to have recent payment issues.
  <u>4. Account Tenure & Delinquency:</u> Shorter account tenure shows weak to moderate positive correlation with delinquency.

- Unexpected anomalies:
  <u>1. Extremely high-income customers with missed payments:</u> Requires further investigation to

understand underlying causes (e.g., data entry errors or atypical financial behavior).
2. Zero or negative balances paired with high debt utilization: Could indicate inconsistencies in the dataset or special financial arrangements.
3. Sudden jumps in missed payments for stable customers: May reflect seasonal or temporary financial stress, worth deeper analysis.

5. AI & GenAI Usage

Generative AI tools were used to summarize the dataset, impute missing data, and detect patterns. AI prompts included:

- "Summarize key patterns, outliers, and missing values in this dataset. Highlight any fields that might present problems for modeling delinquency."
- "Suggest an imputation strategy for missing values in this dataset based on industry best practices."
- "Examine this dataset for delinquency risk. Summarize key correlations between relevant variables and highlight any unusual data points in 3–5 bullet points under 'Correlations' and 'Anomalies.'"

6. Conclusion & Next Steps

Key factors linked to lower delinquency risk include high credit scores, low debt-to-income ratios, moderate credit utilization, and longer account tenure. Data anomalies such as extreme or negative values need verification before modeling.

Next steps:

1. Clean and standardize the dataset, including anomalies and inconsistent categorical labels.
2. Handle missing values using median imputation for numeric fields and a "Missing/Unknown" category for monthly payment statuses.
3. Engineer relevant features to enhance predictive power.
4. Build and validate predictive models using identified risk indicators.
5. Continuously monitor unusual patterns to ensure model reliability.